WASTEWATER MANAGEMENT

Standard Operating Procedure (SOP) Reference Pack

How real company SOPs are structured, controlled, used, and linked to OpsLens

IMPORTANT STATUS
This is an illustrative reference and document-ingestion test pack, not a real company's approved SOP manual. All procedures, limits, setpoints, response times, chemical instructions, PPE, and escalation paths must be reviewed and approved by the site's responsible engineers, EHS team, and operations management before operational use.

Prepared for: OpsLens industrial HMI copilot development
Version: 0.1 • Date: 29 October 2026
Document class: Training / prototype reference • Approval: NOT APPROVED

1. Purpose and scope

This pack demonstrates how wastewater-treatment organizations commonly structure controlled SOPs and how those documents can be organized for reliable retrieval by OpsLens. It includes a document-control model, a representative wastewater process overview, and example procedure templates covering common operational situations.

- Intended audience: project developers, process engineers, operators, maintenance, EHS, document control, and QA reviewers.
- Intended use: UI prototyping, PDF ingestion tests, retrieval/citation validation, and preparation for site-specific SOP authoring.
- Not intended for: direct operation of a plant, selecting chemical doses, setting process limits, bypassing interlocks, or replacing approved site procedures.

2. How SOPs are actually managed in a company

In a well-controlled organization, an SOP is a formally identified, reviewed, approved, version-controlled document. The authoritative copy is normally managed in a document-control system, quality-management system, EHS platform, or controlled document repository. Operators access the current approved revision; superseded revisions are retained for records but are not used as current instructions.

Control field	Typical company practice	Why OpsLens needs it
Document ID	Unique code such as WWTP-OPS-012 (example only)	Stable citation and lookup key
Title / process area	Clear title and plant/unit scope	Find the right procedure
Owner	Responsible department or role	Routes review and maintenance
Revision	Controlled revision number	Avoid mixing instructions from revisions
Status	Draft, In Review, Approved, Superseded, Obsolete	Only approved and applicable content can be authoritative
Approval record	Named role/person, date, controlled approval record	Approval must be verifiable, not inferred
Effective date	Date approved revision becomes effective	Select the applicable revision
Review date	Scheduled review or review trigger	Identify overdue review
Applicability	Plant, line, equipment, process, operating mode	Prevent using the wrong SOP
Training / competency	Required training or authorization	Confirms users are qualified where policy requires
Related documents	Risk assessment, permit, isolation procedure, forms	Preserve dependencies and safety controls
Change history	What changed, why, who approved	Auditability

Important distinction
A PDF is a file format, not proof of approval. A document becomes authoritative only when its identity, revision, approval status, effective applicability, and controlled source are verified.

3. Typical wastewater-treatment process areas

Process area	Common operational purpose	Example records / tags
Inlet works / screening	Remove coarse debris and monitor screen restriction	Differential pressure, screen status, inlet flow
Equalization	Buffer flow/load variation	Tank level, inlet/outlet flow, mixer status
pH adjustment / dosing	Maintain site-defined chemistry using approved controls	pH, dosing-system status, chemical inventory
Biological treatment / aeration	Support biological treatment under site-specific operating conditions	Dissolved oxygen, blower status, basin conditions
Clarification / settling	Separate solids from treated water	Sludge blanket, turbidity, scraper status
Filtration / disinfection	Additional treatment where installed	Differential pressure, turbidity, disinfectant residual
Sludge handling	Collect, condition, thicken, dewater, and manage sludge	Pump status, tank level, equipment status
Final effluent monitoring	Verify discharge quality against permit/site	Flow, pH, turbidity and regulated

	limits	parameters

Not every facility uses every process area. Actual process diagrams, permitted discharge conditions, equipment manuals, risk assessments, and approved operating limits must define the site-specific system.

4. Common sections in a controlled SOP

- Document-control block: ID, title, revision, status, owner, approver, effective date, and review date.
- Purpose and scope: what process/equipment the procedure covers and when it applies.
- Responsibilities and authorization: operator, shift supervisor, maintenance, laboratory, EHS, and escalation roles.
- Prerequisites: required training, permits, risk assessment, PPE, tools, and required process conditions.
- Hazards and controls: chemical, biological, electrical, mechanical, confined-space, pressure, and environmental risks as applicable.
- Procedure: numbered steps with decision points, warnings, expected observations, and stop/escalation criteria.
- Acceptance criteria: site-approved limits and how measurements are verified.
- Abnormal conditions and emergency response: actions defined by approved site procedures and emergency plans.
- Records: log sheets, samples, alarm history, inspection records, work orders, and shift handover.
- References and attachments: process diagrams, equipment manuals, permits, forms, and linked controlled documents.
- Revision history and approvals: traceable changes and authorization.

5. Example SOP document-control lifecycle

1. Author drafts a procedure with process engineering and operations input.
2. Relevant stakeholders review technical accuracy, hazards, environmental requirements, and interfaces.
3. Authorized approvers approve the revision under the company's document-control rules.
4. Document control publishes the effective revision and marks previous revisions superseded.
5. Affected personnel receive required training or communication.
6. The SOP is reviewed periodically and after significant process, equipment, incident, or regulatory changes.
7. Records preserve the approval trail and the historical revision used at a particular time.

6. Example procedure templates

Use limitation
The examples below are intentionally generic. They illustrate realistic document structure and the type of information an industrial SOP contains. They are not step-by-step instructions for operating a real wastewater plant. Replace every placeholder using site-approved engineering documents before operational use.

WWTP-OPS-001 — Routine Inlet Works and Screening Inspection

Document-control field	Example value
Document ID	WWTP-OPS-001 (illustrative)
Revision / status	Rev 0.1 / DRAFT — NOT APPROVED
Owner / approver	Site Operations / designated authorized approver — to be assigned
Effective / review date	Not assigned; complete through formal document control
Applicability	Inlet channel, screen, differential-pressure instrumentation, and associated flow measurement; exact equipment IDs to be listed by the site.

Purpose. Provide a controlled framework for recording the condition of inlet screening equipment and recognizing abnormal restriction or flow conditions.

Prerequisites and controls. Use the site's approved inspection, PPE, access, isolation, and guarding requirements. Do not reach into or remove guarding from operating machinery.

Procedure outline (requires site authoring and approval):

8. Confirm the equipment and tag identities against the current process diagram and equipment register.
9. Review current alarms and recent recorded readings; verify instrument status and freshness.
10. Perform only the visual checks and operator observations authorized by the approved site procedure.
11. Record observed screen condition, flow behavior, differential-pressure indication, and any instrument fault.
12. Compare results with the site's approved operating limits and escalation criteria.
13. If intervention, isolation, or maintenance is required, follow the separate approved procedure and authorization process.
14. Record the inspection result and communicate outstanding issues at shift handover.

Stop and escalate if: Unexpected rise in restriction, abnormal flow, unsafe access conditions, instrument disagreement, overflow risk, or any condition outside approved limits.

Records to retain. Inspection checklist, relevant readings and timestamps, alarm/event IDs, defect/work-order reference, operator identity, and handover notes.

WWTP-OPS-002 — Abnormal Differential-Pressure Alarm Assessment

Document-control field	Example value
Document ID	WWTP-OPS-002 (illustrative)
Revision / status	Rev 0.1 / DRAFT — NOT APPROVED
Owner / approver	Site Operations / designated authorized approver — to be assigned
Effective / review date	Not assigned; complete through formal document control
Applicability	Screen, filter, strainer, or other equipment with differential-pressure monitoring; applicability must identify the exact equipment.

Purpose. Define how an operator records and assesses a differential-pressure alarm without assuming the cause from the alarm alone.

Prerequisites and controls. Use the approved alarm response matrix, alarm priority definitions, instrumentation procedure, and site escalation rules.

Procedure outline (requires site authoring and approval):

15. Confirm the alarm ID, equipment, alarm timestamp, configured threshold, and current alarm state.
16. Check whether the pressure readings are current, valid, correctly scaled, and associated with the correct tags.
17. Review recent differential-pressure trend and related flow readings over the site's defined context window.
18. Check for instrument faults, maintenance activity, recent process changes, and contradictory evidence.
19. Compare observations with the current approved alarm response procedure and its revision.
20. Record the evidence and uncertainty. Do not label a suspected restriction as a confirmed cause without sufficient verification.
21. Follow the approved response and escalation path; do not bypass an alarm or interlock.

Stop and escalate if: High-high or trip alarm, rapidly worsening trend, overflow risk, unreliable instrumentation, conflicting measurements, or unknown safe response.

Records to retain. Alarm ID, source severity, threshold/revision, readings with timestamps, trend window, related events, operator assessment, and escalation record.

WWTP-OPS-003 — Influent Flow Reduction Assessment

Document-control field	Example value
Document ID	WWTP-OPS-003 (illustrative)
Revision / status	Rev 0.1 / DRAFT — NOT APPROVED
Owner / approver	Site Operations / designated authorized approver — to be assigned
Effective / review date	Not assigned; complete through formal document control
Applicability	Inlet flow measurement and associated upstream equipment, where installed.

Purpose. Provide a structured assessment of a recorded reduction in influent flow and its possible relationship to upstream restrictions or measurement faults.

Prerequisites and controls. Use the current process flow diagram, instrument validation procedure, approved alarm matrix, and site-specific response criteria.

Procedure outline (requires site authoring and approval):

22. Confirm the flow tag, engineering units, timestamp, data quality, and configured alarm limits.
23. Compare the current reading with previous valid readings over the approved analysis window.
24. Review related upstream level, screen differential-pressure, pump status, and valve feedback only where these records exist.
25. Check for maintenance, planned shutdown, process changes, or instrument-quality flags.
26. Document plausible explanations separately from confirmed observations.
27. Escalate according to the approved site response matrix when limits or operating conditions require it.

Stop and escalate if: Flow outside permitted operating conditions, overflow or loss-of-containment risk, invalid measurements, or an unexplained sustained excursion.

Records to retain. Flow readings and timestamps, related tag readings, instrument status, alarm IDs, operating context, and action/escalation record.

WWTP-OPS-004 — pH Monitoring and Abnormal Reading Escalation

Document-control field	Example value
Document ID	WWTP-OPS-004 (illustrative)
Revision / status	Rev 0.1 / DRAFT — NOT APPROVED
Owner / approver	Site Operations / designated authorized approver — to be assigned
Effective / review date	Not assigned; complete through formal document control
Applicability	pH measurement and any installed chemical dosing system; exact process and chemicals must be specified by the site.

Purpose. Define a document structure for monitoring pH measurements and escalating abnormal or unreliable readings.

Prerequisites and controls. Use approved chemical-handling procedures, safety data sheets, dosing-system instructions, PPE requirements, and site-approved pH limits. This template does not specify chemical quantities or dosing instructions.

Procedure outline (requires site authoring and approval):

28. Confirm the pH tag, sampling point, timestamp, instrument status, and approved operating range.
29. Check calibration or instrument-quality status according to the approved instrument procedure.
30. Where required, obtain a confirmation sample using the approved sampling procedure and authorized personnel.
31. Compare the result with site limits and evaluate whether the reading is credible and current.
32. Follow the approved site response and escalation procedure for abnormal pH.
33. Record any approved intervention using the required controlled work and chemical-handling records.

Stop and escalate if: Reading outside approved limits, suspected chemical release, unsafe exposure, instrument disagreement, or an unknown dosing-system condition.

Records to retain. pH values, sample ID and time, calibration status, instrument alarms, applicable limits/revision, and escalation records.

WWTP-OPS-005 — Aeration System Alarm and Performance Review

Document-control field	Example value
Document ID	WWTP-OPS-005 (illustrative)
Revision / status	Rev 0.1 / DRAFT — NOT APPROVED
Owner / approver	Site Operations / designated authorized approver — to be assigned
Effective / review date	Not assigned; complete through formal document control
Applicability	Biological treatment basin, blower, aeration equipment, and installed instrumentation.

Purpose. Provide a controlled framework for reviewing an aeration-system alarm and related process measurements.

Prerequisites and controls. Use approved equipment manuals, electrical/mechanical safety procedures, process operating limits, and authorized maintenance rules.

Procedure outline (requires site authoring and approval):

34. Confirm the alarm, equipment ID, operating state, and timestamp.
35. Review available blower status, dissolved-oxygen readings, process indicators, and instrument-quality flags.
36. Distinguish a process condition from an equipment fault or sensor fault using available evidence.
37. Check whether the alarm is associated with maintenance or a known operating-mode change.
38. Follow the approved alarm response and maintenance escalation procedure.
39. Record the assessment and any unresolved process risk for the responsible shift or engineering team.

Stop and escalate if: Loss of required treatment capability, equipment overheating, electrical/mechanical hazard, process limits exceeded, or unavailable critical instrumentation.

Records to retain. Alarm/event IDs, equipment status, available process readings, instrument status, work-order references, and handover notes.

WWTP-OPS-006 — Final Effluent Monitoring and Nonconformance Escalation

Document-control field	Example value
Document ID	WWTP-OPS-006 (illustrative)
Revision / status	Rev 0.1 / DRAFT — NOT APPROVED
Owner / approver	Site Operations / designated authorized approver — to be assigned

Effective / review date	Not assigned; complete through formal document control
Applicability	Final effluent sampling point, flow measurement, laboratory results, and applicable discharge permit.

Purpose. Define a document structure for recording final-effluent monitoring results and escalating potential nonconformance.

Prerequisites and controls. Use the facility's current permit, approved sampling plan, laboratory methods, chain-of-custody process, and reporting requirements.

Procedure outline (requires site authoring and approval):

40. Verify the sampling point, sample identifier, collection time, and required sampling method.
41. Record field readings and laboratory results with units, methods, and quality-control status.
42. Compare results only with the applicable approved permit or site limits and their effective version.
43. If a result is suspect, follow the approved confirmation and laboratory quality procedure.
44. Escalate a confirmed or suspected nonconformance using the site's environmental reporting and response process.
45. Retain traceable records and document required notifications according to the permit and company procedure.

Stop and escalate if: Potential permit exceedance, invalid sample, loss of required monitoring, or a discharge condition requiring immediate escalation under the site plan.

Records to retain. Sample ID, collection time, chain of custody, analytical method, results and units, quality-control record, applicable permit limit, and notifications.

WWTP-OPS-007 — Simulated Alarm Recovery and Source-Clear Verification

Document-control field	Example value
Document ID	WWTP-OPS-007 (illustrative)
Revision / status	Rev 0.1 / DRAFT — NOT APPROVED
Owner / approver	Site Operations / designated authorized approver — to be assigned
Effective / review date	Not assigned; complete through formal document control
Applicability	OpsLens Scenario Lab and test-only incident lifecycle; not a plant-control procedure.

Purpose. Define how a software test environment validates recovery logic without confusing simulated readings with actual plant telemetry.

Prerequisites and controls. Use an isolated test database and explicitly configured simulation mode. Confirm that no PLC write or live-control connection is enabled.

Procedure outline (requires site authoring and approval):

46. Select a documented test scenario and record its initial conditions.
47. Trigger the supported simulation action and confirm the generated reading's source is marked simulated.
48. Verify the reading is persisted with the correct tag, equipment, value, and timestamp.
49. Check the configured clear condition using deterministic backend logic.
50. Verify that the incident remains active when the clear condition is not satisfied.
51. Verify that the incident transitions only when the configured valid clear condition is satisfied.
52. Confirm the frontend reflects the backend state and that the test leaves production records untouched.

Stop and escalate if: Any live-control connection, incorrect equipment mapping, unexpected database mutation, bypassed interlock, or resolution without a valid clear condition.

Records to retain. Scenario ID, initial state, generated reading, source flag, timestamps, lifecycle transitions, API responses, and test result.

7. Example controlled forms and records

Record	Typical fields
Operator shift log	Date/shift, operator, equipment, observations, readings, alarms, actions, outstanding issues
Alarm response record	Alarm ID, timestamp, priority, acknowledgment, evidence reviewed, escalation, outcome
Sampling record	Sample ID, location, collection time, method, sampler, preservation/handling, laboratory result
Maintenance/work order	Asset ID, defect, risk controls, authorization, work performed, verification, return-to-service approval
SOP change request	Document ID, reason, risk/impact assessment, reviewers, approvals, effective revision
Incident review	Event timeline, confirmed facts, hypotheses, root-cause method, corrective actions, owner, due date
Training record	Procedure revision, training date, personnel, assessment/authorization where required

8. How OpsLens should ingest and use these documents

53. Ingest the controlled source document and preserve the original file, document ID, revision, status, and page references.
54. Extract text while retaining headings, numbered steps, warnings, tables, and page references where possible.
55. Flag scanned or incomplete documents for OCR or manual review; never silently treat missing text as a complete procedure.
56. Split content into meaningful sections and steps without separating warnings from the instructions they qualify.
57. Index chunks in the existing retrieval system with source document, revision, approval, equipment applicability, section, step, and page metadata.
58. At query time, retrieve only applicable content and validate the document's current approval and revision before presenting it as authoritative.
59. Show exact source citations in the UI and open the correct PDF page or verified extracted section.
60. Keep AI explanations distinct from quoted or summarized approved instructions.
61. If no applicable approved passage is found, still explain the available incident evidence while stating that no approved SOP was retrieved.
62. Log retrieval status, source IDs, and failures so engineering staff can audit why a source was or was not used.

9. Example OpsLens document metadata

The following is a conceptual example for the document library. It is not an actual approved document record.

Field	Example
document_id	example-wwtp-ops-002
sop_id	WWTP-OPS-002
title	Abnormal Differential-Pressure Alarm Assessment
revision	0.1
approval_status	DRAFT_NOT_APPROVED
effective_date	Not assigned
equipment_scope	Must be specified by site owner
source_file	WWTP-OPS-002_DRAFT.docx or approved source PDF when issued
section	6 — Example procedure template
page	Generated page number; validate after export
content_hash	Calculate during ingestion

10. Site approval checklist

- ☐ The process owner confirms the procedure matches the actual plant and equipment.

- ☐ All limits, setpoints, units, timing requirements, and acceptance criteria come from controlled engineering sources.
- ☐ Hazards, PPE, permits, isolation, chemical handling, and emergency instructions are reviewed by competent personnel.
- ☐ Environmental permit and regulatory obligations are checked by the responsible team.
- ☐ The correct approvers authorize the document under company policy.
- ☐ Revision, effective date, applicability, and review date are recorded.
- ☐ Affected personnel receive required training or communication.
- ☐ The approved revision is published in the controlled document repository.
- ☐ The ingestion pipeline indexes the approved revision and validates citations.
- ☐ OpsLens clearly distinguishes approved instructions, drafts, obsolete versions, and AI-generated hypotheses.

11. Important limitations

This reference pack is designed to help the OpsLens team understand common company SOP structures and test document ingestion and retrieval. It is not based on a particular company's controlled document system and must not be represented as an actual approved plant manual. The example document IDs and process descriptions are illustrative. Actual operation must follow the facility's approved SOPs, equipment manuals, risk assessments, permits, and emergency procedures.

For the OpsLens demo
Use this pack to test document parsing, section/step citations, revision metadata, and the UI's draft/unapproved state. To demonstrate approved guidance, obtain a genuine approved SOP from the responsible wastewater-treatment facility and ingest it with verifiable approval metadata.